

Deepraj Pandey

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EDUCATION

ASHOKA UNIVERSITY

POST GRADUATE DIPLOMA IN
ADVANCED STUDIES AND RESEARCH IN
COMPUTER SCIENCE
May 2021 | Sonapat, India

ASHOKA UNIVERSITY

BSC. HONS. IN COMPUTER SCIENCE
May 2020 | Sonapat, India

D.A.V. PUBLIC SCHOOL

May 2017 | Bhubaneswar, India

SKILLS

PROGRAMMING

Current:

Typescript • C++ • Python • C • Javascript
• Dart • Flutter • Vue • Hyperledger
Fabric • $\text{\LaTeX}$

Familiar:

Bash/ZSH • PowerShell • DevOps •
Android

Other:

Git • Processing • Google Apps Script
Arduino • Ruby on Rails • TI-BASIC

Systems/Simulators:

Linux • MacOS • AWS • GCP • Gem5
NVMain • Champsim • JOS Kernel

COURSEWORK

Programming Language Design (TA)
Advanced Computer Architecture
Program Analysis*
Operating Systems
Computer Security
Algorithm Design
Advanced Programming

* Adv elective in first year.

VOLUNTEERING

Alumni Relations Committee | 2020

Part of the student committee set up to
strengthen student alumni relationships.

Ashoka Student Ambassador | 2019

Student contact for prospective applicants
from Eastern India.

MLH Local Hack Day | Dec, 2017

Organised a 12 hour internal hackathon at
Ashoka University.

MISCELLANEOUS

Calligraphy
Rollerskates
Mechanical Keyboard Enthusiast

PROJECTS

KEY VALUE STORE ON INTEL SGX | [GITHUB LINK](#)

Aug - Dec 2020

- Designed and developed a basic in-memory key value store on Intel SGX.
- Language: C++; followed TDD, ensuring memory safety using Valgrind during development.
- Part of a larger project on exploring feasible KVS architecture (based on works like **ShieldStore**) on Intel SGX. [Link to a presentation on the work.](#)

DEAD BLOCK CORRELATING PREFETCHERS ON CHAMPSIM | [GITHUB LINK](#)

Jan - May 2020

- Reproduceability study of the DBCP paper from 2021 with proposed improvements.
- Language: C++, C, Bash, Python; Checked the performance of the prefetcher on more recent workloads (SPEC 2017).
- [Link to the report.](#)

SPLIDWISE: FINANCE MANAGEMENT DAPP | [GITHUB LINK](#)

Jan - May 2020

- Designed and led development of a distributed application to help users split their bills by recording payments.
- Tech Stack: **Hyperledger Fabric 1.4**, NodeJS REST API with OAuth, VueJS, AWS, Certbot SSL.
- [Links to Report, Presentation.](#)

EXPERIENCE

DEPARTMENT OF COMPUTER SCIENCE | TEACHING ASSISTANT

August 2020 - Present | Ashoka University, India

- TA under **Prof. Gautam Paul** for **Introduction to Computer Programming**, Spring 2021.
- TA under **Prof. Manu Awasthi** for **Programming Language Design and Implementation**, Monsoon 2020.
- Handled course logistics, designed & graded assignments and exams.
- Conducted weekly office hours to cover supplementary material.

BLOCKCHAIN RESEARCH GROUP | RESEARCH ASSISTANT

May 2020 - September 2020

- Worked with **Prof. Mahavir Jhawar** and built **CovidBloc** - a transparent, privacy-preserving contact tracing system on blockchain.
- Submitted to **SPACE 2020** under short paper category. [[eprint](#)]
- Implemented a framework based on the **Google-Apple Exposure Notifications Crypto Specification** from scratch in Dart and TypeScript.
- Built the EN server on Hyperledger Fabric, the web dashboard for health officials on Vue (improved framework allows for **offline verification** of patients), & Android app on Flutter (partial contribution).

DEPARTMENT OF COMPUTER SCIENCE | RESEARCH ASSISTANT

May 2019 - Aug 2019 | Ashoka University, India

- Worked with **Prof. Manu Awasthi** on non-volatile memory architecture in hand-held devices.
- Designed an interface to **gem5** to add **arbitrarily deep** cache hierarchies with non-volatile memory (via NVMain) for simulations.
- Code reviewed and added to an in-house memory simulation tool.

PROJECTS

Superhuman Browser Extension | Freelance

Reverse engineered Superhuman Email Client & added a feature to send automatic text messages on email opens. Private freelance project.

Garbage Collector | GitHub

Implemented the mark and sweep phases of a stop the world garbage collector for Programming Language Design and Implementation class.

Gina | GitHub

A bot built on first principles for syntax analysis and to learn a new language based on user translations.

DES Implementation | GitHub

A DES implementation in **C++** from scratch.

Network Design Cisco Packet Tracer

Designed multiple networks on Cisco Packet Tracer - inter-networks with DHCP servers, web hosting, DNS servers, VLAN, NAT enabled routers (private networks) etc.

OS Labs on xv6

Coursework similar to MIT 6.828. Edited kernel code to add physical page allocators, interrupt handlers, system calls, page faults, and context switches.

Trace Driven Cache Simulator | GitHub

Simple, proof of work trace based cache simulator with 32KB L1 caches (4 words).

Distributed Server | GitHub

Distributed, multithreaded TCP server with a periodic concurrency phase that ensures all servers have the same copy of the data sent from clients.

COMMUNITY

Computer Architecture Reading Group - Summer '19, Spring '20

Google Developer Group, Bhubaneswar: 2015 - Present

Google Developer Day: 2016 (Bhubaneswar), 2019 (Gurgaon)

LINKS

Github:// [DeeprajPandey](#)
LinkedIn:// [DeeprajPandey](#)
Academia:// [DeeprajPandey](#)
Twitter:// [@DeeprajPandey](#)
dblp:// [DeeprajPandey](#)

CONFERENCES

INTERNATIONAL SYMPOSIUM FOR COMPUTER ARCHITECTURE

| UNDERGRADUATE WORKSHOP ATTENDEE

June 2019 | Phoenix, USA

- One of 42 undergrads selected from across the world for the **first undergraduate architecture workshop** at ACM FCRC - ISCA 2019.
- Received funding from UC Davis and a travel grant from Ashoka.
- Mentee under **Dr. Parthasarathy Ranganathan** as a part of the Senior Architect Mentor Matching program at the workshop.

MLH HACKCON VI | GITHUB SCHOLAR

June 2018 | Pennsylvania, USA

- Awarded the GitHub Scholarship for attending the student hackathon organiser conference organised by MLH and GitHub.
- Only student organiser representing a high school hackathon (HackDAV).
- Ideated a hacker exchange program with UMass & UTexas with GitHub.

ISTEM HACKATHON | WINNER

Jan 2019 | IIIT Bangalore, India

- Worked in a team of VI and non-VI participants to build web app that extracts text data from coding tutorial videos for viewers with visual impairments.
- Our team won the hackathon which had 75+ participants including Graduate students and professionals from across the country.

NOTRE DAME LEADERSHIP SEMINARS | PARTICIPANT

July 2016 | Notre Dame, USA

- Only student from India (among the 98 from across the world) selected for a **fully-funded** summer program at the University of Notre Dame.
- Seminar on the ethical concerns of space exploration under **Prof. Don Howard** and **Dr. Jessica Baron**.

LEADERSHIP

REDBRICK HACKS | CO-HEAD

August 2020 - Present | Ashoka University, India

- Currently co-heading the organising team of Ashoka University's flagship hackathon.
- Managing a team of **40** undergraduate students across **6** departments: finance, tech, marketing, hacker relations, legal, events.

WOMEN IN COMPUTING SOCIETY | HEAD OF EVENTS

August 2019 - July 2020 | Ashoka University, India

- Initially shared the position, later headed a team of 8. Organised workshops, guest talks, and socials, usually around inclusivity in Tech.
- Took workshops on Git, Docker, Intro to Programming, web technologies since 2017 on behalf of the society.

MLH SUMMER LEAGUE HACKATHONS | MENTOR & JUDGE

May 2020 - August 2020 | Remote

- Judged **20-30** submissions at every hackathon in the MLH Summer League - one online event **every weekend** all summer.
- **Mentored** multiple projects at the hackathons that had 500-700 participants from all over the world.

HACKDAV | FOUNDER

October 2015 - Present | Bhubaneswar, India

- Founded the state's first hackathon for high school students.
- Applicants from across India and 5 other countries.
- Led the team that raised **INR 18,00,000** in sponsorships for 2017.
- Major sponsors include DigitalOcean, Soylent, JetBrains, MLH, Balsamiq, Hackerearth along with Google and Microsoft Developer Groups.